

Agreement Theorems and Deference*

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1 INTRODUCTION

Two people agree to disagree about a proposition if they commonly know each other's credences in that proposition but their credences differ. Aumann's (1976) *Agreement Theorem* says that if you and I share a prior and have partitional evidence, we cannot agree to disagree.

Two people agree to disagree about an estimate of the value of some quantity if they commonly know each other's estimates but their estimates differ. Sebenius & Geanakoplos (1983) extend Aumann's result: we cannot agree to disagree about estimates either.

Both theorems assume that you and I have precise credences—that your credences and mine are each represented by a single probability function. But many think that rational credences are often imprecise—represented by a set of probability functions, a *representor*, rather than a single one.¹ What becomes of the agreement theorems in this setting?

We can agree to disagree. If you and I share a prior representor, we can commonly know that our posterior imprecise credences and imprecise estimates differ. But there is a limit to how much we can disagree. I prove a result I call the *Overlap Theorem*: our commonly known posterior imprecise credences and imprecise estimates must overlap.²

Both agreement theorems also assume partitional evidence. This assumption is controversial: partitionality entails that you always know exactly what your

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1 For defenses of imprecise credences, see Levi (1974), Jeffrey (1983), van Fraassen (1990), Walley (1991), Kaplan (1996), Schoenfield (2012), Sturgeon (2010), and Joyce (2010). For challenges to imprecise credences, see White (2010), Elga (2010), and Schoenfield (2017).

2 Kajii & Ui (2005) were the first to prove overlap, for credences under partitional evidence.

evidence is. But many think this is implausible: our information-gathering processes are fallible, and when they fail, they do not always tell us that they have. Geanakoplos (1989) shows that Aumann's Theorem requires less: a weaker condition, *balancedness*, suffices for agreement. He also introduces a strengthening of balancedness, *positive balancedness*. I show that positive balancedness suffices for overlap.

Balancedness and positive balancedness look quite unlike familiar constraints on evidence. Your evidence is balanced on a proposition when the indicator function of that proposition is a weighted sum of the indicator functions of the bodies of evidence you might have when that proposition is true. Positive balancedness adds only that the weights are nonnegative. Why should conditions of this form determine whether two people can agree to disagree? Both look like technical conditions that happen to make the proofs go through—conditions with no epistemic significance of their own.

They are nothing of the kind. I show that each is equivalent to a principle of deference. Balancedness is equivalent to *Total Certainty*: if you are certain what your better-informed self's estimate of some quantity is, yours should match it. Positive balancedness is equivalent to *Total Containment*: if you are certain that estimate her falls within some interval, yours should fall in that interval too. So Aumann's Agreement Theorem follows from Total Certainty, and the Overlap Theorem from Total Containment. Total Containment also yields an agreement result for precise estimates: two people who satisfy Total Containment cannot commonly know that their estimates fall in disjoint intervals. This holds whether or not either knows the other's exact estimate.

The equivalences have a further consequence: they show when we *can* agree to disagree. Suppose you violate Total Certainty on some proposition that is self-evident to you—one you know whenever it is true. Then, by a corollary of the first equivalence, you can agree to disagree with someone who satisfies Total Certainty. The second equivalence yields a parallel conclusion about Total Containment. Suppose you violate Total Containment on some proposition self-evident to you. Then there is someone who satisfies Total Containment on every proposition self-evident to her and with whom you can commonly know that your estimates lie in disjoint intervals.³

Together, these results establish the paper's central claim: deference is what stands in the way of agreeing to disagree, and it is all that stands in the way. Those who defer cannot agree to disagree with one another, and anyone who does not defer can agree to disagree with someone who does.

³ Whether the derivation of overlap from Total Containment has a converse—whether someone who fails Total Containment can commonly know, with someone who satisfies it, that their imprecise estimates are disjoint—I leave open.

The role of deference in agreement theorems is not confined to probabilistic settings. Consider comparative confidence judgments—judgments of the form *I am at least as confident in p as in q* . Can you and I commonly know that I am at least as confident in p as in q , while you are not? To answer this, I work within a standard framework in which your comparative confidence judgments are represented by a confidence ordering.⁴ A confidence ordering may be complete, ranking every pair of propositions, or incomplete, leaving some pairs incomparable. I show that when our orderings are complete, we cannot agree to disagree about how p and q rank. This follows from *Comparative Deference*: if you are certain how your better-informed self ranks p against q , you should rank them the same way. When our orderings are incomplete, we can agree to disagree about how p and q rank, but only if at least one of us judges them incomparable.

Deference explains one more result. Sebenius & Geanakoplos (1983) show that two people who share a prior and have partitional evidence cannot commonly know that each strictly prefers her side of a bet to declining it. I show that Total Containment suffices for this result too.

The paper proceeds as follows. §2 presents the precise agreement theorems and introduces balancedness and positive balancedness. §3 moves to imprecise credences, showing that those with imprecise credences can agree to disagree and proving the Overlap Theorem. §4 derives agreement and overlap from principles of deference, establishes the equivalences—balancedness with Total Certainty, and positive balancedness with Total Containment—and shows that those who fail to defer can agree to disagree with those who do defer. §5 extends the analysis to comparative confidence. §6 takes up betting. §7 concludes.

2 THE AGREEMENT THEOREMS

I begin in §2.1 by introducing the framework within which we will state and prove the agreement theorems. In §2.2, I prove Aumann’s Agreement Theorem, along with Sebenius & Geanakoplos’s extension of it to expectations, under partitional evidence. In §2.3, I prove the same results with balancedness in place of partitionality.

2.1 Preliminaries

A *precise agreement frame* is a structure:⁵

$$\mathcal{F} = \langle \Omega, E_A, E_B, \pi_A, \pi_B, p_A, p_B \rangle$$

⁴ My framework largely follows Eva & Stern (2023).

⁵ For simplicity I restrict attention to two people. Every result holds for any finite number of people.

Ω is a finite set of states. E_i is i 's *evidence function*: $E_i(w)$ is i 's evidence at w . π_i is i 's *prior function*: $\pi_i(w)$ is i 's prior at w . p_i is i 's *posterior function*: $p_i(w)$ is i 's posterior at w .

Aumann imposes three conditions on evidence.⁶

Truth

$$w \in E_i(w)$$

Positive Introspection

$$\text{If } w' \in E_i(w), \text{ then } E_i(w') \subseteq E_i(w)$$

Negative Introspection

$$\text{If } w' \in E_i(w), \text{ then } E_i(w) \subseteq E_i(w')$$

Together, these three conditions make your evidence *partitional*: $\{E_i(w) : w \in \Omega\}$ is a partition of Ω , where $E_i(w)$ is your evidence cell containing w .

Aumann assumes that A and B have the same prior, and that it does not vary from world to world.⁷

Shared Prior

$$\pi_A(w) = \pi_B(w)$$

Constant Prior

$$\pi_i(w) = \pi_i(w')$$

Write π for this shared prior. I also assume that for all $w \in \Omega$:

Regularity

$$\pi(w) > 0$$

A and B update by conditioning their shared prior on their evidence.

Conditionalization

$$p_i(w) = \pi(\cdot | E_i(w))$$

⁶ Since Truth, Positive Introspection, and Negative Introspection hold at every world in Ω , A and B commonly know that they both satisfy all three conditions. Lederman (2015) shows that this assumption of common knowledge is not idle: those who satisfy the three conditions only at a pointed frame's designated state can agree to disagree, even with a shared prior.

⁷ The names for these theses come from Lederman (2015).

That fixes A's and B's credences at each world. The agreement theorems concern what they commonly know about those credences. Say that i *knows* H at w if $E_i(w) \subseteq H$, and that A and B *commonly know* H at w if they both know H at w , both know that they know H at w , and so on.⁸ This iterative definition of common knowledge has an equivalent formulation that I will use throughout the paper. Say that a proposition H is *self-evident* to i if $E_i(w) \subseteq H$ for all $w \in H$: whenever H is true, i knows it. Say that H is *public* between A and B if it is self-evident to both. Then:

Common Knowledge

A and B commonly know H at w if and only if there is some public F with $w \in F \subseteq H$.⁹

2.2 Agreement with Partitionality

Write $[p_i(H) = x]$ for the proposition that i 's posterior credence in H is x :

$$[p_i(H) = x] = \{w \in \Omega : p_i(w)(H) = x\}$$

Theorem 2.1 (Aumann (1976)). *Let $\mathcal{F}$ be a precise agreement frame satisfying the conditions of §2.1. Let H be a proposition. Let $w \in \Omega$. Let $x, y \in [0, 1]$. Suppose A and B commonly know $[p_A(H) = x]$ and $[p_B(H) = y]$ at w . Then $x = y$.*

The proofs of this theorem and the next rely on the following lemma, whose proof is routine.

Lemma 2.2. *Let $\mathcal{E}$ be a nonempty, finite set of nonempty, pairwise disjoint propositions, and let X be a random variable. Suppose π is regular and that $\mathbb{E}_\pi[X|E] = a$ for every $E \in \mathcal{E}$. Then $\mathbb{E}_\pi[X|\bigcup \mathcal{E}] = a$.*

Proof of Theorem 2.1. Since $[p_A(H) = x]$ and $[p_B(H) = y]$ are commonly known at w , there is a public F with

$$w \in F \subseteq [p_A(H) = x] \cap [p_B(H) = y].$$

So A's posterior credence in H is x throughout F , and B's is y . By Conditionalization, for every $v \in F$,

$$\pi(H|E_A(v)) = x \quad \text{and} \quad \pi(H|E_B(v)) = y.$$

⁸ This iterative definition of common knowledge is due to Schiffer (1972). See Fagin et al. (1995).

⁹ The characterization in terms of public propositions is due to Monderer & Samet (1989), who prove the equivalence. A similar idea can be found in Aumann (1976) and Lewis (1969).

Let $\mathcal{E}_A = \{E_A(v) : v \in F\}$ and $\mathcal{E}_B = \{E_B(v) : v \in F\}$. Since $w \in F$, both sets are nonempty. By Truth, each of their members is nonempty. Because A's evidence is partitional and F is self-evident to A, $\mathcal{E}_A$ is a partition of F . Since π is regular and $\mathbb{E}_\pi[1_H|E] = \pi(H|E) = x$ for every $E \in \mathcal{E}_A$, Lemma 2.2 with $X = 1_H$ gives

$$\pi(H|F) = x.$$

Since $\mathcal{E}_B$ is also a partition of F , applying Lemma 2.2 to $\mathcal{E}_B$ in the same way gives

$$\pi(H|F) = y.$$

So $x = y$. □

Aumann's theorem concerns credences in a proposition. Sebenius & Geanakoplos (1983) extend Aumann's result to expectations of random variables. Write $[\mathbb{E}_{p_i}[X] = a]$ for the proposition that i 's posterior expectation of X is a :

$$[\mathbb{E}_{p_i}[X] = a] = \{w \in \Omega : \mathbb{E}_{p_i(w)}[X] = a\}$$

Theorem 2.3 (Sebenius & Geanakoplos (1983)). *Let $\mathcal{F}$ be a precise agreement frame satisfying the conditions of §2.1. Let X be a random variable. Let $w \in \Omega$. Let $a, b \in \mathbb{R}$. Suppose A and B commonly know $[\mathbb{E}_{p_A}[X] = a]$ and $[\mathbb{E}_{p_B}[X] = b]$ at w . Then $a = b$.*

Proof of Theorem 2.3. The proof of Theorem 2.1 goes through with estimates of X in place of credences in H : replace $[p_A(H) = x]$ and $[p_B(H) = y]$ with $[\mathbb{E}_{p_A}[X] = a]$ and $[\mathbb{E}_{p_B}[X] = b]$, and $\pi(H|\cdot)$ with $\mathbb{E}_\pi[X|\cdot]$. Lemma 2.2 is applied to X rather than to 1_H . □

2.3 Agreement with Balancedness

Partitionality has been widely challenged as a constraint on evidence. It entails that you always know exactly what your evidence is. But your evidence can mislead you without telling you that it has. A wall looks red but is in fact white under red lighting. That the wall is red is not part of your evidence. But nothing indicates that there is a problem: it is not part of your evidence that it is not part of your evidence that the wall is red.¹⁰

Geanakoplos (1989) shows that Truth and a weaker condition, balancedness, suffice for agreement. Say that your evidence function Π is *balanced on F* if there are real numbers y_v for each $v \in F$ such that

¹⁰ For versions of this argument, see McDowell (1982), Williamson (2000), and Salow (2019).

$$\mathbf{1}_F = \sum_{v \in F} y_v \mathbf{1}_{\Pi(v)}.$$

Balancedness has an equivalent formulation in terms of conditional probabilities rather than indicator functions.

Proposition 2.4. *Let P be a regular probability function, Π an evidence function satisfying Truth, and F a nonempty proposition. Π is balanced on F if and only if there are real numbers λ_v for each $v \in F$, with $\sum_{v \in F} \lambda_v = 1$, such that*

$$P(\cdot|F) = \sum_{v \in F} \lambda_v P(\cdot|\Pi(v)).$$

Proof. F is nonempty, and by Truth so is $\Pi(v)$ for every $v \in F$. Since P is regular, $P(F) > 0$ and $P(\Pi(v)) > 0$ for every $v \in F$. For any proposition H with $P(H) > 0$ and any $w \in \Omega$,

$$P(w) \mathbf{1}_H(w) = P(H) P(w|H). \quad (\dagger)$$

Fix real numbers y_v for $v \in F$ and set $\lambda_v = y_v P(\Pi(v))/P(F)$. Since $P(w)$, $P(F)$, and each $P(\Pi(v))$ are positive, the following are equivalent for every $w \in \Omega$:

$$\begin{aligned} \mathbf{1}_F(w) &= \sum_{v \in F} y_v \mathbf{1}_{\Pi(v)}(w), \\ P(w) \mathbf{1}_F(w) &= \sum_{v \in F} y_v P(w) \mathbf{1}_{\Pi(v)}(w), \\ P(F) P(w|F) &= \sum_{v \in F} y_v P(\Pi(v)) P(w|\Pi(v)), \\ P(w|F) &= \sum_{v \in F} \lambda_v P(w|\Pi(v)). \end{aligned}$$

The second line follows from the first by multiplying both sides by $P(w)$. The third uses $(\dagger)$, with $H = F$ on the left and $H = \Pi(v)$ on the right. Dividing by $P(F)$ gives the fourth. Each step is reversible. So $\mathbf{1}_F = \sum_v y_v \mathbf{1}_{\Pi(v)}$ if and only if $P(\cdot|F) = \sum_v \lambda_v P(\cdot|\Pi(v))$. The map $y_v \mapsto \lambda_v$ is a bijection on $\mathbb{R}$, so Π is balanced on F if and only if $P(\cdot|F) = \sum_{v \in F} \lambda_v P(\cdot|\Pi(v))$ for some real λ_v . $P(\cdot|F)$ and each $P(\cdot|\Pi(v))$ sum to 1 over $w \in \Omega$. Summing the last identity over $w \in \Omega$ therefore gives $\sum_{v \in F} \lambda_v = 1$. $\square$

Geanakoplos also introduces a strengthening of balancedness. Π is *positively balanced on F* if it is balanced on F with every $y_v \geq 0$. Positive balancedness also has an equivalent probabilistic formulation.

Corollary 2.5. *Let P be a regular probability function, Π an evidence function satisfying Truth, and F a nonempty proposition. Π is positively balanced on F if*

and only if there are nonnegative real numbers λ_v for each $v \in F$, with $\sum_{v \in F} \lambda_v = 1$, such that

$$P(\cdot|F) = \sum_{v \in F} \lambda_v P(\cdot|\Pi(v)).$$

Proof of Corollary 2.5. The proof of Proposition 2.4 uses two families of weights, the λ_v and the y_v , related by

$$\lambda_v = \frac{y_v P(\Pi(v))}{P(F)}.$$

For every $v \in F$, $P(\Pi(v))/P(F)$ is strictly positive, so $\lambda_v \geq 0$ if and only if $y_v \geq 0$. $\square$

I will call Geanakoplos's version of balancedness the *indicator formulation* and the version in terms of conditional probabilities the *probabilistic formulation*. I know of no earlier statement of the equivalences between the two. The probabilistic formulations make it easier to see how balancedness might be connected to deference: they relate your credence conditional on F to the credences you might have when F is true. In §4, I use these formulations to prove that balancedness is equivalent to Total Certainty and positive balancedness to Total Containment.

The indicator formulation makes no reference to a probability function. The probabilistic formulation does. Proposition 2.4 therefore has an important consequence: whether the probabilistic formulation holds does not depend on which regular probability function we choose. Given Truth, if Π is balanced on F , then for every regular P there are weights λ_v , summing to one, such that

$$P(\cdot|F) = \sum_{v \in F} \lambda_v P(\cdot|\Pi(v)).$$

The same holds for positive balancedness, by Corollary 2.5. I work with the probabilistic formulations from here on.

We can now replace partitionality with balancedness in the agreement theorems.

Theorem 2.6. *Let $\mathcal{F}$ be a precise agreement frame satisfying the conditions of §2.1 other than Positive and Negative Introspection, with E_A and E_B each balanced on every nonempty public proposition. Let H be a proposition, $w \in \Omega$, and $x, y \in [0, 1]$. Suppose A and B commonly know $[p_A(H) = x]$ and $[p_B(H) = y]$ at w . Then $x = y$.*

Proof of Theorem 2.6. Since $[p_A(H) = x]$ and $[p_B(H) = y]$ are commonly known at w , there is a public proposition F such that

$$w \in F \subseteq [p_A(H) = x] \cap [p_B(H) = y].$$

By Conditionalization,

$$\pi(H|E_A(v)) = x \quad \text{and} \quad \pi(H|E_B(v)) = y$$

for every $v \in F$. Since F is nonempty and public, E_A and E_B are each balanced on F : there are weights λ_v and μ_v , each family summing to one, such that

$$\pi(\cdot|F) = \sum_{v \in F} \lambda_v \pi(\cdot|E_A(v)) = \sum_{v \in F} \mu_v \pi(\cdot|E_B(v)).$$

Evaluating at H gives

$$x = \sum_{v \in F} \lambda_v \pi(H|E_A(v)) = \pi(H|F) = \sum_{v \in F} \mu_v \pi(H|E_B(v)) = y.$$

So $x = y$. □

We now show that agreement about expectations needs only balancedness too.

Theorem 2.7 (Geanakoplos (1989)). *Let $\mathcal{F}$ be a precise agreement frame satisfying the conditions of §2.1 other than Positive and Negative Introspection, with E_A and E_B each balanced on every nonempty public proposition. Let X be a random variable, $\omega \in \Omega$, and $a, b \in \mathbb{R}$. Suppose A and B commonly know $[\mathbb{E}_{p_A}[X] = a]$ and $[\mathbb{E}_{p_B}[X] = b]$ at ω . Then $a = b$.¹¹*

The proof uses the following lemma, which I state without proof.

Lemma 2.8. *If $Q = \sum_{j=1}^n \lambda_j Q_j$ for probability functions $Q, Q_1, \dots, Q_n$ on Ω and real numbers $\lambda_1, \dots, \lambda_n$, then $\mathbb{E}_Q[X] = \sum_{j=1}^n \lambda_j \mathbb{E}_{Q_j}[X]$ for every random variable X .*

Proof of Theorem 2.7. Since $[\mathbb{E}_{p_A}[X] = a]$ and $[\mathbb{E}_{p_B}[X] = b]$ are commonly known at ω , there is a public proposition F such that

$$\omega \in F \subseteq [\mathbb{E}_{p_A}[X] = a] \cap [\mathbb{E}_{p_B}[X] = b].$$

By Conditionalization, $\mathbb{E}_\pi[X|E_A(v)] = a$ and $\mathbb{E}_\pi[X|E_B(v)] = b$ for every $v \in F$. Since F is nonempty and public, E_A and E_B are each balanced on F : there are weights λ_v and μ_v , each family summing to one, such that

$$\pi(\cdot|F) = \sum_{v \in F} \lambda_v \pi(\cdot|E_A(v)) = \sum_{v \in F} \mu_v \pi(\cdot|E_B(v)).$$

By Lemma 2.8,

$$a = \sum_{v \in F} \lambda_v \mathbb{E}_\pi[X|E_A(v)] = \mathbb{E}_\pi[X|F] = \sum_{v \in F} \mu_v \mathbb{E}_\pi[X|E_B(v)] = b.$$

So $a = b$. □

¹¹ Geanakoplos (1989, Theorem 6) takes the states to be real numbers and proves agreement for the random variable $X(\omega) = \omega$. Here X can be any random variable on Ω . His definition of balancedness allows weights on all sets in the range of the evidence function, but his proof restricts them to sets contained in the public proposition. I impose this restriction from the start.

3 AGREEMENT AND IMPRECISION

The agreement theorems of §2 assume precise credences. What happens when credences are imprecise? We will see that even with partitional evidence, a shared prior, and common knowledge of both, two people can agree to disagree about both propositions and estimates.

3.1 Preliminaries

A *representor* is a nonempty, closed, convex set of probability functions on Ω .¹² An *imprecise agreement frame* is a structure

$$\mathcal{F} = \langle \Omega, E_A, E_B, C_A, C_B, c_A, c_B \rangle.$$

Ω is a finite set of states. E_i is i 's evidence function: $E_i(w)$ is i 's evidence at w . C_i is i 's prior representor function: $C_i(w)$ is i 's prior representor at w . c_i is i 's posterior representor function: $c_i(w)$ is i 's posterior representor at w .

A and B share a prior representor, and it doesn't vary from world to world.

Shared Representor

$$C_A(w) = C_B(w)$$

Constant Representor

$$C_i(w) = C_i(w')$$

Write C for this shared prior representor. I assume that every member of C assigns positive probability to every world.

Regularity

$$P(w) > 0 \text{ for all } P \in C$$

Your *imprecise credence* in H is the set of credences in H represented by members of C . Your imprecise credence in H conditional on E is the corresponding set of conditional credences:

$$C(H) = \{P(H) : P \in C\}, \quad C(H|E) = \{P(H|E) : P \in C\}.$$

¹² The proofs of Theorems 3.3 and 3.4 do not require closure or convexity. Given Regularity, these conditions guarantee that $C(H|E)$ and $\mathbb{E}_C[X|E]$ are closed intervals for every nonempty E . Convexity is standard (Levi 1980, Walley 1991).

Your *imprecise expectation* of a random variable X is the set of expectations of X represented by members of C . Your imprecise expectation of X conditional on E is the corresponding set of conditional expectations:

$$\mathbb{E}_C[X] = \{\mathbb{E}_P[X] : P \in C\}, \quad \mathbb{E}_C[X|E] = \{\mathbb{E}_P[X|E] : P \in C\}.$$

A's and B's posterior representors are obtained by conditioning every member of C on their evidence.

Imprecise Conditionalization

$$c_i(w) = \{P(\cdot|E_i(w)) : P \in C\}$$

3.2 Agreeing to Disagree

The next two examples show that A and B can agree to disagree: about a proposition in the first and an estimate in the second. Evidence is partitional in both.¹³

Example 3.1 (Agreeing to Disagree about a Proposition). Let $\Omega = \{w_1, w_2, w_3, w_4\}$ and $H = \{w_1, w_3\}$. A's evidence partitions Ω into $E_1 = \{w_1, w_2\}$ and $E_2 = \{w_3, w_4\}$. B's evidence is Ω at every world. The shared representor is $C = \{\alpha P + (1 - \alpha)Q : \alpha \in [0, 1]\}$, where P and Q assign probabilities as follows.

P			Q		
w_1	w_2	E_1	w_1	w_2	E_1
0.4	0.1		0.1	0.4	
w_3	w_4	E_2	w_3	w_4	E_2
0.1	0.4		0.4	0.1	
H	$\neg H$		H	$\neg H$	

Since $P(H) = Q(H) = 0.5$, every member of C assigns H probability 0.5. B's evidence is Ω at every world, so $c_B(w)(H) = \{0.5\}$ for every $w \in \Omega$.

Consider A. Take any $R = \alpha P + (1 - \alpha)Q \in C$. Since $P(E) = Q(E) = 0.5$ for each $E \in \{E_1, E_2\}$, $R(\cdot|E) = \alpha P(\cdot|E) + (1 - \alpha)Q(\cdot|E)$. Now $P(H|E_1) = Q(H|E_2) = 0.8$ and $P(H|E_2) = Q(H|E_1) = 0.2$, so

$$R(H|E_1) = 0.2 + 0.6\alpha \quad \text{and} \quad R(H|E_2) = 0.8 - 0.6\alpha.$$

¹³ The first example is adapted from Carvajal & Correia-da-Silva (2013). See also Zhang, Liu, & Seidenfeld (2018, Example 2), whose set of priors is neither convex nor regular.

As α ranges over $[0, 1]$, $R(H|E_1)$ and $R(H|E_2)$ each range over $[0.2, 0.8]$. So $c_A(w)(H) = [0.2, 0.8]$ for every $w \in \Omega$.

A's and B's posteriors in H are constant throughout Ω , so both are commonly known. Yet $[0.2, 0.8] \neq \{0.5\}$. A and B agree to disagree.

Imprecision also permits commonly known disagreement about expectations.

Example 3.2 (Agreeing to Disagree about an Estimate). Let $\Omega = \{w_1, \dots, w_6\}$, and let X take the value 10 at w_1 and w_4 , 5 at w_2 and w_5 , and 0 at w_3 and w_6 . A's evidence partitions Ω into $E_1 = \{w_1, w_2, w_3\}$ and $E_2 = \{w_4, w_5, w_6\}$. B's evidence is Ω at every world. The shared representor is $C = \{\alpha P + (1 - \alpha)Q : \alpha \in [0, 1]\}$, where P and Q assign probabilities as follows.

P				Q			
w_1	w_2	w_3	E_1	w_1	w_2	w_3	E_1
0.3	0.1	0.1		0.1	0.1	0.3	
w_4	w_5	w_6	E_2	w_4	w_5	w_6	E_2
0.1	0.1	0.3		0.3	0.1	0.1	
$X = 10$	$X = 5$	$X = 0$		$X = 10$	$X = 5$	$X = 0$	

Since $\mathbb{E}_P[X] = \mathbb{E}_Q[X] = 5$, Lemma 2.8 gives $\mathbb{E}_R[X] = 5$ for every $R \in C$. B's evidence is Ω at every world, so $\mathbb{E}_{c_B(w)}[X] = \{5\}$ for every $w \in \Omega$.

Consider A. Take any $R = \alpha P + (1 - \alpha)Q \in C$. As in Example 3.1, $P(E) = Q(E) = 0.5$ for each $E \in \{E_1, E_2\}$, so $R(\cdot|E) = \alpha P(\cdot|E) + (1 - \alpha)Q(\cdot|E)$. Now $\mathbb{E}_P[X|E_1] = \mathbb{E}_Q[X|E_2] = 7$ and $\mathbb{E}_P[X|E_2] = \mathbb{E}_Q[X|E_1] = 3$, so by Lemma 2.8

$$\mathbb{E}_R[X|E_1] = 3 + 4\alpha \quad \text{and} \quad \mathbb{E}_R[X|E_2] = 7 - 4\alpha.$$

As α ranges over $[0, 1]$, $\mathbb{E}_R[X|E_1]$ and $\mathbb{E}_R[X|E_2]$ each range over $[3, 7]$. So $\mathbb{E}_{c_A(w)}[X] = [3, 7]$ for every $w \in \Omega$.

A's and B's posterior expectations are constant throughout Ω , so both are commonly known. Yet $[3, 7] \neq \{5\}$. A and B agree to disagree.

3.3 The Overlap Theorem

Kajii & Ui (2005) prove that, for two people who share a set of priors and have partitional evidence, commonly known posterior credal intervals must overlap.¹⁴

¹⁴ Carvajal & Correia-da-Silva (2013) prove the result without assuming a shared representor. They assume only that A's and B's prior representors are closed, connected, and intersecting.

I strengthen that result in two ways. Commonly known posterior estimates must overlap too. I also show that neither overlap result requires partitional evidence: positive balancedness suffices.

Write $[c_i(H) = I]$ for the proposition that i 's imprecise posterior in H is I :

$$[c_i(H) = I] = \{w \in \Omega : c_i(w)(H) = I\}$$

Theorem 3.3 (Kajii & Ui (2005)). *Let $\mathcal{F}$ be an imprecise agreement frame satisfying the conditions of §3.1, with E_A and E_B partitional. Let H be a proposition. Let $w \in \Omega$. Let I_A and I_B be closed subintervals of $[0, 1]$. Suppose $[c_A(H) = I_A]$ and $[c_B(H) = I_B]$ are both commonly known at w . Then*

$$I_A \cap I_B \neq \emptyset.$$

If your evidence is partitional, it is positively balanced on every nonempty proposition self-evident to you.¹⁵ The following theorem replaces partitionality with positive balancedness and extends overlap from credences in propositions to expectations of arbitrary random variables. Theorem 3.3 follows by setting $X = 1_H$.

Theorem 3.4 (Overlap for Estimates). *Let $\mathcal{F}$ be an imprecise agreement frame satisfying Truth and the conditions of §3.1, with E_A and E_B each positively balanced on every nonempty public proposition. Let X be a random variable, $w \in \Omega$, and I_A and I_B closed intervals. Suppose A and B commonly know $[\mathbb{E}_{c_A}[X] = I_A]$ and $[\mathbb{E}_{c_B}[X] = I_B]$ at w . Then $I_A \cap I_B \neq \emptyset$.*

Proof. There is a public F with $w \in F$ throughout which A 's imprecise posterior expectation of X is I_A and B 's is I_B . Fix any $P \in C$. By Imprecise Conditionalization,

$$\mathbb{E}_P[X|E_A(v)] \in I_A \quad \text{and} \quad \mathbb{E}_P[X|E_B(v)] \in I_B$$

for every $v \in F$. Since F is nonempty and public, E_A and E_B are each positively balanced on F : there are nonnegative weights λ_v and μ_v , each family summing to one, such that

$$P(\cdot|F) = \sum_{v \in F} \lambda_v P(\cdot|E_A(v)) = \sum_{v \in F} \mu_v P(\cdot|E_B(v)).$$

By Lemma 2.8, $\mathbb{E}_P[X|F]$ is a weighted average of the $\mathbb{E}_P[X|E_A(v)]$, each of which lies in I_A , so $\mathbb{E}_P[X|F] \in I_A$. The same argument for B gives $\mathbb{E}_P[X|F] \in I_B$. So $I_A \cap I_B \neq \emptyset$. $\square$

¹⁵ Let Π be partitional and F nonempty and self-evident to Π . Then $\{\Pi(v) : v \in F\}$ is a partition of F , so for any regular P ,

$$P(\cdot|F) = \sum_{v \in F} \frac{P(v)}{P(F)} P(\cdot|\Pi(v)),$$

with nonnegative coefficients summing to one.

4 DEFERENCE AND AGREEMENT

This section connects agreement and overlap to deference to your better-informed self. I derive agreement from one deference principle—*Total Certainty*—and overlap from another—*Total Containment*, in a form suited to imprecise credences. I then show that, on any nonempty proposition self-evident to you, balancedness is equivalent to Total Certainty and positive balancedness to Total Containment. The equivalences have an important consequence: someone who fails to defer on a proposition self-evident to her can agree to disagree with someone who does.

4.1 *Certainty and Containment*

Let P be your credence function over Ω . Assume P is regular. Let Π be the evidence function of your better-informed self, and suppose it satisfies Truth. Her credence function is yours conditioned on what she has learned: $P(\cdot|\Pi(w))$. Write $[P(H|\Pi) = x]$ for the proposition that her credence in H is x :

$$[P(H|\Pi) = x] = \{w \in \Omega : P(H|\Pi(w)) = x\}.$$

Certainty says that if you are certain that your better-informed self assigns H credence x , you should assign H credence x too.

Certainty

If $P([P(H|\Pi) = x]) = 1$, then $P(H) = x$.

Certainty tells you to defer to your better-informed self's credences. You should also defer to her estimates: if you are certain what her estimate of X is, your own estimate of X should match it. Write $[\mathbb{E}_P[X|\Pi] = a]$ for the proposition that her estimate of X is a :

$$[\mathbb{E}_P[X|\Pi] = a] = \{w \in \Omega : \mathbb{E}_P[X|\Pi(w)] = a\}.$$

Then:

Total Certainty

If $P([\mathbb{E}_P[X|\Pi] = a]) = 1$, then $\mathbb{E}_P[X] = a$.¹⁶

Total Certainty governs your unconditional estimates. You should also defer to your better-informed self when forming conditional estimates: conditional

¹⁶ The label follows Dorst et al. (2021), who put ‘Total’ before the name of a deference principle to mark its extension from credences in propositions to expectations of arbitrary random variables.

on any nonempty proposition F , you should defer to your better-informed self's estimates conditional on F . Conditional on F , your credence function is $P(\cdot|F)$, and your better-informed self's credence function at ν is $P(\cdot|\Pi(\nu) \cap F)$. This is defined whenever $P(\Pi(\nu) \cap F) > 0$. By Truth, $\nu \in \Pi(\nu) \cap F$ for every $\nu \in F$, so by Regularity it is defined at every $\nu \in F$. Write $[\mathbb{E}_P[X|\Pi \cap F] = a]$ for the proposition that her estimate of X conditional on F is a :

$$[\mathbb{E}_P[X|\Pi \cap F] = a] = \{\omega \in \Omega : \Pi(\omega) \cap F \neq \emptyset \text{ and } \mathbb{E}_P[X|\Pi(\omega) \cap F] = a\}.$$

Then:

Total Certainty on F

If $P([\mathbb{E}_P[X|\Pi \cap F] = a]|F) = 1$, then $\mathbb{E}_P[X|F] = a$.

When I speak of Total Certainty without qualification, I mean the principle on every nonempty F . The principles below are stated on F in the same way, and I take each to hold for every nonempty F .

Total Certainty on F tells you what your estimate should be when you know your better-informed self's estimate exactly. But often you are certain only that her estimate falls within some range. *Total Containment on F* says your estimate should fall within that range too. Write $[\mathbb{E}_P[X|\Pi \cap F] \in [a, b]]$ for the proposition that her estimate of X conditional on F lies in $[a, b]$. Then:

Total Containment on F

If $P([\mathbb{E}_P[X|\Pi \cap F] \in [a, b]]|F) = 1$, then $\mathbb{E}_P[X|F] \in [a, b]$.

Now suppose your credences are imprecise, represented by a representor C every member of which is regular. Your better-informed self's imprecise estimate of X conditional on F at ν is $\mathbb{E}_C[X|\Pi(\nu) \cap F]$. Write $[\mathbb{E}_C[X|\Pi \cap F] \subseteq [a, b]]$ for the proposition that it lies in $[a, b]$. Then:

Imprecise Total Containment on F

If $C([\mathbb{E}_C[X|\Pi \cap F] \subseteq [a, b]]|F) = \{1\}$, then $\mathbb{E}_C[X|F] \subseteq [a, b]$.

The next section derives agreement and overlap from principles of deference. Overlap is a claim about representors, so we need a principle of deference for representors. Imprecise Total Containment on F serves this purpose. It follows from Total Containment on F : if each member of C satisfies Total Containment on

F , then C satisfies Imprecise Total Containment on F .¹⁷ When I say that overlap follows from Total Containment, this is what I mean.

4.2 Deriving Agreement and Overlap

Consider a precise agreement frame satisfying the conditions of §2.1 other than Positive and Negative Introspection. Write π for the shared prior. Let X be a random variable, $w \in \Omega$, and $a, b \in \mathbb{R}$. Suppose A and B commonly know $[\mathbb{E}_{p_A}[X] = a]$ and $[\mathbb{E}_{p_B}[X] = b]$ at w , so that there is a public F with $w \in F \subseteq [\mathbb{E}_{p_A}[X] = a] \cap [\mathbb{E}_{p_B}[X] = b]$. We show that Total Certainty on this F yields $a = b$.

Since F is public, $E_A(v) \subseteq F$ at every $v \in F$, so $E_A(v) \cap F = E_A(v)$. By Conditionalization,

$$\mathbb{E}_{p_A(v)}[X] = \mathbb{E}_\pi[X|E_A(v)] = \mathbb{E}_\pi[X|E_A(v) \cap F]$$

for every $v \in F$. Since A's estimate is a throughout F , $\pi([\mathbb{E}_\pi[X|E_A \cap F] = a]|F) = 1$. By Total Certainty on F with $\Pi = E_A$, $\mathbb{E}_\pi[X|F] = a$. The same reasoning for B gives $\mathbb{E}_\pi[X|F] = b$. So $a = b$.

Say that two people *agree to disagree in intervals* when they commonly know that their estimates lie in disjoint closed intervals. Total Containment entails that A and B cannot agree to disagree in intervals. Suppose A and B commonly know at w that their estimates of X lie in $[a, a']$ and $[b, b']$, respectively. Choose a public F containing w throughout which A's estimate lies in $[a, a']$ and B's in $[b, b']$. Since F is self-evident to A, we have $E_A(v) \cap F = E_A(v)$ at every $v \in F$. By Conditionalization,

$$\pi([\mathbb{E}_\pi[X|E_A \cap F] \in [a, a']]|F) = 1.$$

By Total Containment on F , with $\Pi = E_A$, it follows that $\mathbb{E}_\pi[X|F] \in [a, a']$. By applying the same argument to B, we conclude that $\mathbb{E}_\pi[X|F] \in [b, b']$. Both intervals contain $\mathbb{E}_\pi[X|F]$, so A and B cannot agree to disagree in intervals.

Now consider an imprecise agreement frame satisfying Truth and the conditions of §3.1. Write C for the shared representor. Let X be a random variable, $w \in \Omega$, and I_A and I_B closed intervals. Suppose A and B commonly know $[\mathbb{E}_{c_A}[X] = I_A]$ and $[\mathbb{E}_{c_B}[X] = I_B]$ at w , so that there is a public F with $w \in F \subseteq [\mathbb{E}_{c_A}[X] = I_A] \cap [\mathbb{E}_{c_B}[X] = I_B]$. We show that Imprecise Total Containment on this F yields $I_A \cap I_B \neq \emptyset$.

Since F is public, $E_A(v) \subseteq F$ at every $v \in F$, so $E_A(v) \cap F = E_A(v)$. By Imprecise Conditionalization,

$$\mathbb{E}_{c_A(v)}[X] = \mathbb{E}_C[X|E_A(v)] = \mathbb{E}_C[X|E_A(v) \cap F]$$

¹⁷ Suppose $C([\mathbb{E}_C[X|\Pi \cap F] \subseteq [a, b]]|F) = \{1\}$, and fix $P \in C$. Then $P([\mathbb{E}_C[X|\Pi \cap F] \subseteq [a, b]]|F) = 1$. Since $\mathbb{E}_P[X|\Pi(v) \cap F] \in \mathbb{E}_C[X|\Pi(v) \cap F]$ for every $v \in F$, it follows that $P([\mathbb{E}_P[X|\Pi \cap F] \subseteq [a, b]]|F) = 1$. By Total Containment on F , $\mathbb{E}_P[X|F] \in [a, b]$. Since P was arbitrary, $\mathbb{E}_C[X|F] \subseteq [a, b]$.

for every $v \in F$. Since A's imprecise estimate is I_A throughout F , $C([\mathbb{E}_C[X|E_A \cap F] \subseteq I_A] \mid F) = \{1\}$. By Imprecise Total Containment on F with $\Pi = E_A$, $\mathbb{E}_C[X \mid F] \subseteq I_A$. The same reasoning for B gives $\mathbb{E}_C[X \mid F] \subseteq I_B$. Since C is nonempty, so is $\mathbb{E}_C[X \mid F]$, and therefore $I_A \cap I_B \neq \emptyset$.

4.3 Equivalences

We now show that, for any nonempty F self-evident to Π , balancedness on F is equivalent to Total Certainty on F , and positive balancedness on F to Total Containment on F .¹⁸

Fix a regular probability function P , an evidence function Π satisfying Truth, and a nonempty proposition F self-evident to Π .¹⁹ Recall that Π is balanced on F if

$$P(\cdot \mid F) = \sum_{v \in F} \lambda_v P(\cdot \mid \Pi(v))$$

for weights λ_v summing to 1, and positively balanced on F if in addition every $\lambda_v \geq 0$.

Since F is self-evident to Π , $\Pi(v) \subseteq F$ for every $v \in F$, so $\Pi(v) \cap F = \Pi(v)$. Since P is regular, $P(\cdot \mid F)$ assigns positive probability to every world in F . So a proposition has probability 1 conditional on F if and only if it is true throughout F . The antecedent of Total Certainty on F therefore says that $\mathbb{E}_P[X \mid \Pi(v)] = a$ at every $v \in F$, and the antecedent of Total Containment on F that $\mathbb{E}_P[X \mid \Pi(v)] \in [a, b]$ at every $v \in F$. We use these formulations in the proofs below.

Theorem 4.1. Π is balanced on F if and only if Total Certainty on F holds.²⁰

- 18 Zhang (2026) uses the label ‘Total Certainty’ for a different principle. Her *Weak Deference* says that if you are certain that the expert’s credence function belongs to a set Γ , then your own credence function must belong to the convex hull of Γ . In my setting, where the expert is your better-informed self, this is equivalent to positive balancedness on Ω . Her Fact 10 shows Weak Deference equivalent to a special case of her Total Certainty. This closely parallels my Theorem 4.2 in the case $F = \Omega$. We arrived at these results independently.
- 19 The proofs also work without self-evidence: for any nonempty F , replace $\Pi(v)$ with $\Pi(v) \cap F$, and the same arguments show that Total Certainty on F holds if and only if $v \mapsto \Pi(v) \cap F$ is balanced on F , and Total Containment on F if and only if it is positively balanced. Schultheis & Zhang (ms) call the positive condition *conditional positive balancedness*, and show that Total Containment holds on every nonempty F if and only if conditional positive balancedness does. I keep the restriction to self-evident F so as to retain Geanakoplos’s formulation of balancedness.
- 20 Certainty on F (unlike Total Certainty on F) does not imply balancedness on F . Let $\Omega = \{1, 2, 3\}$, $F = \Omega$, $P = (0.2, 0.5, 0.3)$, $\Pi(1) = \{1, 2\}$, and $\Pi(2) = \Pi(3) = \{2, 3\}$. The two posteriors are

$$P(\cdot \mid \{1, 2\}) = (2/7, 5/7, 0) \quad \text{and} \quad P(\cdot \mid \{2, 3\}) = (0, 5/8, 3/8).$$

Proof. ($\Rightarrow$) Suppose Π is balanced on F . Then there are weights λ_ν summing to 1 such that

$$P(\cdot|F) = \sum_{\nu \in F} \lambda_\nu P(\cdot|\Pi(\nu)).$$

Let X be any random variable. By Lemma 2.8,

$$\mathbb{E}_P[X|F] = \sum_{\nu \in F} \lambda_\nu \mathbb{E}_P[X|\Pi(\nu)].$$

Suppose $\mathbb{E}_P[X|\Pi(\nu)] = a$ for every $\nu \in F$. Then

$$\mathbb{E}_P[X|F] = \sum_{\nu \in F} \lambda_\nu a = a,$$

so Total Certainty on F holds.

($\Leftarrow$) Suppose Total Certainty on F holds. Suppose, for contradiction, that Π is not balanced on F . Let

$$S = \{P(\cdot|\Pi(\nu)) : \nu \in F\}$$

and write $q = P(\cdot|F)$. Regard each probability distribution over Ω as a vector in $\mathbb{R}^{|\Omega|}$. Balancedness on F says that q is an affine combination of members of S —that q lies in the affine hull of S . Since Π is not balanced on F , q lies outside the affine hull of S .

Fix $s_0 \in S$, and let L be the subspace spanned by the differences $s - s_0$ for $s \in S$. Then $\text{aff}(S) = s_0 + L$. Since $q \notin \text{aff}(S)$, $q - s_0 \notin L$. Every vector in $\mathbb{R}^{|\Omega|}$ is the sum of a vector in L and a vector orthogonal to L . Write

$$q - s_0 = \ell + z,$$

where $\ell \in L$ and z is orthogonal to L . Since $q - s_0 \notin L$, $z \neq 0$.

For every $s \in S$, $s - s_0 \in L$. So $z \cdot (s - s_0) = 0$, which gives

$$z \cdot s = z \cdot s_0.$$

Let $c = z \cdot s_0$. The dot product with z takes the value c at every member of S . But $\ell \in L$, so $z \cdot \ell = 0$, and

$$z \cdot (q - s_0) = z \cdot (\ell + z) = z \cdot z > 0.$$

They assign the same probability to H only when $H = \emptyset$ or $H = \Omega$. In both cases, $P(H|F)$ agrees, so Certainty on F holds. But balancedness on F would require

$$P = \lambda(2/7, 5/7, 0) + (1 - \lambda)(0, 5/8, 3/8).$$

The second coordinate forces $\lambda = -7/5$, which makes the first coordinate $-2/5$ rather than $1/5$. So Π is not balanced on F .

So $z \cdot q \neq c$.

Define a random variable Z by $Z(w) = z_w$. For any probability distribution μ ,

$$z \cdot \mu = \sum_{w \in \Omega} z_w \mu(w) = \mathbb{E}_\mu[Z].$$

Since the members of S are the $P(\cdot|\Pi(v))$ for $v \in F$, and $q = P(\cdot|F)$, it follows that $\mathbb{E}_P[Z|\Pi(v)] = c$ for every $v \in F$, while $\mathbb{E}_P[Z|F] \neq c$. This contradicts Total Certainty on F , so Π is balanced on F . $\square$

Theorem 4.2. Π is positively balanced on F if and only if Total Containment on F holds.

Proof. ($\Rightarrow$) Suppose Π is positively balanced on F . Then there are nonnegative weights λ_v summing to 1 such that

$$P(\cdot|F) = \sum_{v \in F} \lambda_v P(\cdot|\Pi(v)).$$

Let X be any random variable. By Lemma 2.8,

$$\mathbb{E}_P[X|F] = \sum_{v \in F} \lambda_v \mathbb{E}_P[X|\Pi(v)].$$

Suppose $\mathbb{E}_P[X|\Pi(v)] \in [a, b]$ for every $v \in F$. Since the λ_v are nonnegative and sum to 1, $\mathbb{E}_P[X|F]$ is a weighted average of numbers in $[a, b]$, so $\mathbb{E}_P[X|F] \in [a, b]$ and Total Containment on F holds.

($\Leftarrow$) Suppose Total Containment on F holds. Suppose, for contradiction, that Π is not positively balanced on F . Let

$$S = \{P(\cdot|\Pi(v)) : v \in F\}$$

and write $q = P(\cdot|F)$. Positive balancedness on F says that q is a convex combination of members of S —that q lies in the convex hull of S . Since Π is not positively balanced on F , q lies outside the convex hull of S . Since S is finite, its convex hull is closed.

By the separating hyperplane theorem, choose $z \in \mathbb{R}^{|\Omega|}$ and $t \in \mathbb{R}$ such that

$$z \cdot q < t \leq z \cdot s$$

for every $s \in S$. Define a random variable Z by $Z(w) = z_w$. As in the proof of Theorem 4.1, $z \cdot \mu = \mathbb{E}_\mu[Z]$ for every probability distribution μ . So

$$\mathbb{E}_P[Z|F] < t \leq \mathbb{E}_P[Z|\Pi(v)]$$

for every $v \in F$. Let $b = \max_{v \in F} \mathbb{E}_P[Z|\Pi(v)]$. Then $\mathbb{E}_P[Z|\Pi(v)] \in [t, b]$ for every $v \in F$, while $\mathbb{E}_P[Z|F] \notin [t, b]$. This contradicts Total Containment on F . So Π is positively balanced on F . $\square$

Theorem 4.2 also yields Imprecise Total Containment on F —the principle used to derive overlap in §4.2. Fix a representor C whose members are all regular.

Corollary 4.3. *If Π is positively balanced on F , then Imprecise Total Containment on F holds.*

Proof. As noted in §2.3, positive balancedness on F does not depend on which regular probability function we use. So Theorem 4.2 gives Total Containment on F for every $P \in C$. If Total Containment on F holds for each member of C , then Imprecise Total Containment on F holds for C . $\square$

4.4 Disagreement without Deference

We have seen that principles of deference preclude agreeing to disagree. We now turn to the converse. Suppose your evidence is not balanced on a nonempty proposition F self-evident to you. Then there is someone with partitional evidence with whom you can agree to disagree about some quantity throughout F . If your evidence is not positively balanced on F , there is someone with partitional evidence with whom you can agree to disagree in intervals throughout F .

Corollary 4.4. *Let P be a regular probability function, Π an evidence function satisfying Truth, and F a nonempty proposition self-evident to Π . Let $\Pi'(w)$ be F if $w \in F$ and $\Omega \setminus F$ otherwise. Then Π' is partitional, satisfies Truth, and F is self-evident to Π' . Moreover:*

- (i) *if Π is not balanced on F , there is a random variable Z and a real number c such that, for every $v \in F$,*

$$\mathbb{E}_P[Z|\Pi(v)] = c \quad \text{and} \quad \mathbb{E}_P[Z|\Pi'(v)] \neq c.$$

- (ii) *if Π is not positively balanced on F , there is a random variable Z and a closed interval $[a, b]$ such that, for every $v \in F$,*

$$\mathbb{E}_P[Z|\Pi(v)] \in [a, b] \quad \text{and} \quad \mathbb{E}_P[Z|\Pi'(v)] \notin [a, b].$$

Proof. By construction, Π' is partitional. Since $w \in \Pi'(w)$ for every w , Π' satisfies Truth. Since $\Pi'(v) = F$ throughout F , F is self-evident to Π' .

Suppose first that Π is not balanced on F . By Theorem 4.1, Total Certainty on F fails: there is a random variable Z and a real number c such that $\mathbb{E}_P[Z|\Pi(v)] = c$ for all $v \in F$, while $\mathbb{E}_P[Z|F] \neq c$. Since $\Pi'(v) = F$ for all $v \in F$, $\mathbb{E}_P[Z|\Pi'(v)] = \mathbb{E}_P[Z|F] \neq c$.

Now suppose that Π is not positively balanced on F . By Theorem 4.2, Total Containment on F fails: there is a random variable Z and a closed interval

$[a, b]$ such that $\mathbb{E}_P[Z|\Pi(\nu)] \in [a, b]$ for every $\nu \in F$, while $\mathbb{E}_P[Z|F] \notin [a, b]$. Since $\Pi'(\nu) = F$ for every $\nu \in F$, $\mathbb{E}_P[Z|\Pi'(\nu)] = \mathbb{E}_P[Z|F] \notin [a, b]$. $\square$

We can now show:

Proposition 4.5. *Fix a regular prior and suppose your evidence satisfies Truth.*

- (i) *If you fail Total Certainty on some nonempty proposition self-evident to you, there is someone who shares your prior and satisfies Total Certainty on every nonempty proposition self-evident to her with whom you can agree to disagree.*
- (ii) *If you fail Total Containment on some nonempty proposition self-evident to you, there is someone who shares your prior and satisfies Total Containment on every nonempty proposition self-evident to her with whom you can agree to disagree in intervals.*

Proof. (i) Write P for your prior. Let F be a nonempty proposition self-evident to you on which Total Certainty fails. By Theorem 4.1, your evidence function Π is not balanced on F . As in Corollary 4.4, let $\Pi'(w)$ be F if $w \in F$ and $\Omega \setminus F$ otherwise. The corollary gives a random variable Z and a real number c such that

$$\mathbb{E}_P[Z|\Pi(\nu)] = c \quad \text{and} \quad \mathbb{E}_P[Z|\Pi'(\nu)] \neq c$$

for every $\nu \in F$.

Consider someone whose evidence function is Π' and who shares your prior. Since $\Pi'(\nu) = F$ for every $\nu \in F$, her estimate of Z is the constant $d = \mathbb{E}_P[Z|F] \neq c$ throughout F . Since F is self-evident to both of you, it is public, so at every world in F you commonly know that your estimate of Z is c and hers is d . You can agree to disagree. Her evidence is partitional, hence balanced on every nonempty proposition self-evident to her, so by Theorem 4.1 she satisfies Total Certainty on every such proposition.

(ii) Repeat the argument with Theorem 4.2 in place of Theorem 4.1 and the interval $[a, b]$ from Corollary 4.4 in place of c . Your partner's evidence function is again Π' . At every $\nu \in F$, your estimate of Z lies in $[a, b]$, while hers is the constant $d = \mathbb{E}_P[Z|F] \notin [a, b]$. So at every world in F you commonly know that your estimates lie in the disjoint closed intervals $[a, b]$ and $[d, d]$. You can agree to disagree in intervals. $\square$

Proposition 4.5(ii) concerns precise estimates commonly known to lie in disjoint intervals. It is the converse of the result in §4.2 that two people who satisfy Total Containment cannot agree to disagree in intervals—not of the result that commonly known imprecise estimates must overlap when both people satisfy

Imprecise Total Containment. I leave open whether this overlap result has a converse.

Together with §4.2, Proposition 4.5 casts Total Certainty and Total Containment in a new light. Though each appears to be no more than a constraint on how your credences relate to those of your better-informed self, each also settles who can agree to disagree with whom. Among those who share a prior, those who satisfy Total Certainty on every nonempty proposition self-evident to them cannot agree to disagree with one another, and anyone who does not can agree to disagree with someone who does. Likewise for Total Containment and agreeing to disagree in intervals.

Much now rides on Total Certainty and Total Containment. How plausible are they? One reason for doubt: in §2 we rejected partitionality, and deference without partitionality is a delicate matter. Reflection, for example, holds when evidence is partitional and can fail when it is not.²¹ Is the case against Reflection also a case against Total Certainty and Total Containment?

No. Dorst et al. (2021), building on Dorst (2020), show that a weaker deference principle, Total Trust, is consistent with failures of partitionality. Total Trust says: conditional on your better-informed self's estimate of X being at least t , your own estimate of X should be at least t . Total Containment and Total Certainty follow from Total Trust. And there is good reason to accept Total Trust. Dorst et al. (2021, Theorem 2.2) prove it equivalent to Value—the principle that, for any decision you face, you should prefer to leave the choice to your better-informed self. Value has a strong claim to acceptance. Whatever else it involves, deferring to your better-informed self means at least being willing to act on her opinions. Value asks that and nothing more.

5 COMPARATIVE AGREEMENT

Aumann's theorem says that, for those who share a precise prior, commonly known posteriors must coincide. The Overlap Theorem says that, for those who share a representor, commonly known posterior intervals must overlap. We can also ask about agreeing to disagree in non-probabilistic settings.²² Consider *comparative confidence judgments*—judgments of the form *I am at least as confident in p as I am in q* . Can you and I commonly know that I am at least as confident in p as in q while you are not?

21 For discussions of Reflection's failure with nonpartitional evidence, see Williamson (2000, ch. 10), Dorst (2020), and Dorst et al. (2021).

22 Cave (1983) and Bacharach (1985) prove an agreement theorem for decisions satisfying the sure-thing principle. The comparative confidence results are new.

I take up this question within a standard framework for comparative confidence.²³ The framework does not presuppose that we have credences of any kind—precise or imprecise. Instead, each person has a *confidence ordering*: $p \succeq q$ means that you are at least as confident in p as in q . An ordering is *complete* if it ranks every pair and *incomplete* if it leaves some pairs incomparable.²⁴ Completeness is the comparative analogue of precision, incompleteness of imprecision.

Commonly known comparative disagreement is constrained in much the same way as commonly known probabilistic disagreement. Two people who share a prior ordering and update on their evidence cannot commonly know each other's rankings of a pair of propositions when those rankings differ, unless at least one of them judges that pair incomparable. This result, too, follows from a principle of deference—*Comparative Deference*, formulated in entirely comparative terms.

5.1 Confidence Orderings

From a confidence ordering $\succeq$, we derive three relations. You are equally confident in p and q ($p \sim q$) when $p \succeq q$ and $q \succeq p$. You are strictly more confident in p than in q ($p \succ q$) when $p \succeq q$ but not $q \succeq p$. You judge p and q incomparable ($p \odot q$) when neither $p \succeq q$ nor $q \succeq p$. Following standard practice, I require every confidence ordering to satisfy the following three conditions.

Partial Preorder

$\succeq$ is reflexive and transitive.

Non-Triviality

$\top \succ \perp$.

Monotonicity

If $p \models q$, then $q \succeq p$: you are at least as confident in p 's logical consequences as you are in p .

23 I follow the presentation in Eva & Stern (2023). For more on comparative confidence, see Fine (1973), Halpern (2003), and Konek (2019).

24 For arguments against completeness, see Keynes (1921), Kaplan (1996), and Eva & Stern (2023), who call it *Opinionation*.

I also assume de Finetti's (1937) *Quasi-Additivity* axiom.²⁵

Quasi-Additivity

If $r \cap t = s \cap t = \emptyset$, then $r \succeq s$ if and only if $r \cup t \succeq s \cup t$.

Quasi-Additivity says that if you are at least as confident that the die lands one as that it lands six, you should be at least as confident that it lands one or two as that it lands six or two. Quasi-Additivity holds for $\succ$ too: if $r \cap t = s \cap t = \emptyset$, then $r \succ s$ if and only if $r \cup t \succ s \cup t$.²⁶

For any proposition E , let $\succeq^E$ be the conditional ordering defined by

$$p \succeq^E q \quad \text{if and only if} \quad (E \cap p) \succeq (E \cap q).$$

We derive $\sim^E$, $\succ^E$, and $\odot^E$ from $\succeq^E$ just as $\sim$, $\succ$, and $\odot$ were derived from $\succeq$. Note that $\succeq^E$ is reflexive, transitive, monotonic, and quasi-additive. It satisfies Non-Triviality if and only if $E \succ \perp$.

A probability function P determines a complete confidence ordering: $p \succeq q$ if and only if $P(p) \geq P(q)$. When $P(E) > 0$, $\succeq^E$ coincides with the ordering determined by $P(\cdot|E)$: $p \succeq^E q$ if and only if $P(p|E) \geq P(q|E)$.

A representor C determines an ordering that need not be complete: $p \succeq q$ if and only if $P(p) \geq P(q)$ for all $P \in C$, and $p \odot q$ when $P(p) > P(q)$ for some $P \in C$ and $Q(q) > Q(p)$ for some $Q \in C$. When every $P \in C$ assigns positive probability to E , $\succeq^E$ coincides with the ordering determined by $\{P(\cdot|E) : P \in C\}$: $p \succeq^E q$ if and only if $P(p|E) \geq P(q|E)$ for every $P \in C$. Orderings determined by probability functions and representors both satisfy Quasi-Additivity.²⁷

²⁵ Quasi-Additivity is equivalent to

$$p \succeq q \iff (p \setminus q) \succeq (q \setminus p). \quad (*)$$

($\Rightarrow$) Fix p and q , and set $r = p \setminus q$, $s = q \setminus p$, $t = p \cap q$. Then $r \cap t = s \cap t = \emptyset$, $r \cup t = p$, and $s \cup t = q$, so Quasi-Additivity gives (*) directly. ($\Leftarrow$) Suppose $r \cap t = s \cap t = \emptyset$. Then $(r \cup t) \setminus (s \cup t) = r \setminus s$ and $(s \cup t) \setminus (r \cup t) = s \setminus r$. Apply (*) to r, s and again to $r \cup t, s \cup t$. Each of $r \succeq s$ and $r \cup t \succeq s \cup t$ then holds if and only if $(r \setminus s) \succeq (s \setminus r)$. (*) is the formulation Eva & Stern (2023) use.

²⁶ Suppose t is disjoint from r and s . By Quasi-Additivity, $r \succeq s$ if and only if $r \cup t \succeq s \cup t$, and $s \succeq r$ if and only if $s \cup t \succeq r \cup t$. Since $r \succ s$ if and only if $r \succeq s$ and not $s \succeq r$, it follows that $r \succ s$ if and only if $r \cup t \succ s \cup t$.

²⁷ Suppose $r \cap t = s \cap t = \emptyset$. For a single P , finite additivity gives $P(r \cup t) = P(r) + P(t)$ and $P(s \cup t) = P(s) + P(t)$, so $P(r \cup t) \geq P(s \cup t)$ if and only if $P(r) \geq P(s)$. That is the claim for orderings determined by a probability function. For a representor C , apply this to each $P \in C$: $P(r) \geq P(s)$ for every $P \in C$ if and only if $P(r \cup t) \geq P(s \cup t)$ for every $P \in C$.

5.2 Comparative Agreement Frames

A *comparative agreement frame* is a structure:

$$\langle \Omega, E_A, E_B, \succeq, \succeq_A, \succeq_B \rangle$$

Ω is a finite set of worlds. E_i is i 's evidence function. $\succeq$ is A's and B's shared prior ordering.²⁸ $\succeq_i$ is i 's posterior ordering function: $\succeq_i(w)$ is i 's posterior ordering at w . Note that three orderings are in play: the shared prior $\succeq$, i 's posterior $\succeq_i(w)$, and the prior conditional on E , $\succeq^E$.

Following Eva & Stern (2023), I assume that A and B update by comparative conditionalization.

Comparative Conditionalization

$$\succeq_i(w) = \succeq^{E_i(w)}$$

Comparative Conditionalization says that i 's posterior ordering at w is the shared prior ordering conditional on i 's evidence at w . So $p \succeq_i(w)q$ if and only if

$$(E_i(w) \cap p) \succeq (E_i(w) \cap q).$$

5.3 No Comparative Disagreement

We can now state the comparative agreement theorem: you and I cannot commonly know that one of us judges $p \succeq q$ while the other judges $q \succ p$.

Write $[p \succeq_i q]$ for the proposition that i 's posterior ordering ranks p at least as high as q , and $[q \succ_i p]$ for the proposition that it ranks q strictly higher than p :

$$[p \succeq_i q] = \{w \in \Omega : p \succeq_i(w)q\}, \quad [q \succ_i p] = \{w \in \Omega : q \succ_i(w)p\}.$$

Theorem 5.1 (No Comparative Disagreement). *Let*

$$\langle \Omega, E_A, E_B, \succeq, \succeq_A, \succeq_B \rangle$$

be a comparative agreement frame satisfying Comparative Conditionalization. Suppose $\succeq$ satisfies Quasi-Additivity, and suppose that E_A and E_B satisfy Truth, Positive Introspection, and Negative Introspection. Let p and q be propositions. Let $w \in \Omega$. Then $[p \succeq_A q]$ and $[q \succ_B p]$ are not both commonly known at w .

²⁸ The single shared ordering $\succeq$ is the comparative analogue of the Shared Prior and Constant Prior assumptions of §2: A and B start with the same ordering, and it does not vary across worlds.

Proof. Suppose for contradiction that $[p \succeq_A q]$ and $[q \succ_B p]$ are both commonly known at w . Then there is a public proposition F with

$$w \in F \subseteq [p \succeq_A q] \cap [q \succ_B p].$$

So throughout F , A judges $p \succeq q$ and B judges $q \succ p$.

Consider A. At every $v \in F$, A's posterior ordering ranks p at least as high as q . By Comparative Conditionalization, $\succeq_A(v) = \succeq^{E_A(v)}$, so by the definition of $\succeq^E$, $(E_A(v) \cap p) \succeq (E_A(v) \cap q)$. Since E_A is partitional and F is self-evident to A, A's evidence cells within F partition F . Call them $A_1, \dots, A_n$. So $(A_j \cap p) \succeq (A_j \cap q)$ for every j . Define a sequence $G_0, \dots, G_n$ as follows.

$$\begin{aligned} G_0 &= (A_1 \cap p) \cup (A_2 \cap p) \cup \dots \cup (A_n \cap p) = F \cap p, \\ G_1 &= (A_1 \cap q) \cup (A_2 \cap p) \cup \dots \cup (A_n \cap p), \\ &\vdots \\ G_n &= (A_1 \cap q) \cup (A_2 \cap q) \cup \dots \cup (A_n \cap q) = F \cap q. \end{aligned}$$

To pass from G_{j-1} to G_j , replace $A_j \cap p$ with $A_j \cap q$ and leave the terms for all other cells unchanged. Writing $T_j = \bigcup_{k < j} (A_k \cap q) \cup \bigcup_{k > j} (A_k \cap p)$, we have

$$G_{j-1} = T_j \cup (A_j \cap p) \quad \text{and} \quad G_j = T_j \cup (A_j \cap q).$$

Since the cells are disjoint, T_j is disjoint from $A_j \cap p$ and $A_j \cap q$. Since $(A_j \cap p) \succeq (A_j \cap q)$, Quasi-Additivity gives $G_{j-1} \succeq G_j$ for every j . By transitivity of $\succeq$,

$$(F \cap p) \succeq (F \cap q).$$

Now consider B. Since E_B is partitional and F is self-evident to B, B's evidence cells within F partition F . Call them $B_1, \dots, B_m$. As with A, Comparative Conditionalization gives $(B_k \cap q) \succ (B_k \cap p)$ for every k . Define $H_0, \dots, H_m$ from $F \cap q$ to $F \cap p$ in the same way, replacing $B_k \cap q$ with $B_k \cap p$ at step k . Since $(B_k \cap q) \succ (B_k \cap p)$ and Quasi-Additivity holds for $\succ$, $H_{k-1} \succ H_k$ for every k . By transitivity of $\succ$ (which follows from transitivity of $\succeq$),

$$(F \cap q) \succ (F \cap p).$$

This contradicts $(F \cap p) \succeq (F \cap q)$. □

When the shared prior ordering $\succeq$ is complete, Theorem 5.1 rules out every commonly known disagreement. First note that when $\succeq$ is complete, $\succeq^E$ is complete for every E . For any p and q , completeness of $\succeq$ gives either $(E \cap p) \succeq (E \cap q)$ or $(E \cap q) \succeq (E \cap p)$. So either $p \succeq^E q$ or $q \succeq^E p$. By Comparative Conditionalization, A's and B's posterior orderings are complete too. Any disagreement between two

complete orderings has one of the two forms ruled out by Theorem 5.1: one subject judges $p \succeq q$ while the other judges $q \succ p$, or one judges $q \succeq p$ while the other judges $p \succ q$. So if $\succeq$ is complete, A and B cannot commonly know each other's rankings of p and q when those rankings differ.

Call the proof of Theorem 5.1 the *swapping argument*. We move from $F \cap p$ to $F \cap q$ by replacing p with q on one evidence cell at a time. Quasi-Additivity preserves the comparison at each step, and transitivity carries it from $F \cap p$ to $F \cap q$. Can the same argument rule out commonly known disagreement involving incomparability?

5.4 Incomparability

No. The swapping argument needs transitivity. But $\odot$ is not transitive.²⁹ The next example, which has the same structure as Example 3.1, uses the failure of transitivity to produce commonly known comparative disagreement.

Example 5.2 (Comparative Agreeing to Disagree). Let $\Omega = \{w_1, w_2, w_3, w_4\}$, with $p = \{w_1, w_3\}$ and $q = \{w_2, w_4\}$. A's evidence partitions Ω into $E_1 = \{w_1, w_2\}$ and $E_2 = \{w_3, w_4\}$. B's evidence is Ω at every world. The shared ordering $\succeq$ is determined by the representor

$$C = \{\alpha P + (1 - \alpha)Q : \alpha \in [0, 1]\}$$

where P and Q assign probabilities as follows:

P			Q		
w_1	w_2	E_1	w_1	w_2	E_1
0.4	0.1		0.1	0.4	
w_3	w_4	E_2	w_3	w_4	E_2
0.1	0.4		0.4	0.1	
p	q		p	q	

Both P and Q assign probability 0.5 to p and to q . So every member of C does too, and the shared prior ordering ranks p and q equally: $p \sim q$. B's evidence is Ω at every world, so her posterior ordering is always the prior ordering. Now

²⁹ Let $\succeq$ be determined by the representor $C = \{\alpha P + (1 - \alpha)Q : \alpha \in [0, 1]\}$ on $\Omega = \{w_1, w_2, w_3\}$, where $P = (0.5, 0.1, 0.4)$ and $Q = (0.3, 0.5, 0.2)$. Let $p = \{w_1\}$, $q = \{w_2\}$, and $r = \{w_3\}$. P ranks p above q while Q ranks q above p , so $p \odot q$. Q ranks q above r while P ranks r above q , so $q \odot r$. But every member of C ranks p above r , so $p \succ r$.

consider A. Given E_1 , P ranks p above q , while Q ranks q above p . So A judges $p \odot q$ at every world in E_1 . Given E_2 , the rankings reverse, so A judges $p \odot q$ at every world in E_2 . A therefore judges $p \odot q$ throughout Ω , while B judges $p \sim q$ throughout Ω . Both judgments are commonly known. A and B agree to disagree.

A and B can also commonly know that A judges $p \odot q$ while B judges $p \succ q$.

Example 5.3 (Incomparability against a Strict Ranking). Let $P = (0.4, 0.1, 0.2, 0.3)$ and $Q = (0.1, 0.3, 0.5, 0.1)$, with the rest of the construction as in Example 5.2. Both P and Q assign p probability 0.6 and q probability 0.4. So every member of C does too, and the shared prior ordering ranks p strictly above q : $p \succ q$. B learns nothing, so she judges $p \succ q$ at every world. Given E_1 , P ranks p above q , while Q ranks q above p . Given E_2 , these rankings reverse. A therefore judges $p \odot q$ throughout Ω . Both judgments are commonly known. A and B agree to disagree.

Together, No Comparative Disagreement and these two examples settle which comparative disagreements can be commonly known. One of A and B must judge p and q incomparable. There are no further restrictions: they can commonly know that one of them judges p and q incomparable while the other judges $p \sim q$, $p \succ q$, or $q \succ p$.

5.5 Deference and Comparative Agreement

The proof of Theorem 5.1 uses Quasi-Additivity and partitional evidence. But No Comparative Disagreement can also be derived from a principle of deference: *Comparative Deference*. Let $\succeq$ be your confidence ordering, and let Π be the evidence function of your better-informed self, satisfying Truth. Conditional on a nonempty proposition F , her ordering at v is $\succeq^{\Pi(v) \cap F}$.

Comparative Deference on F

For all p and q and every $R \in \{\succ, \sim, \succeq\}$: if $p R^{\Pi(v) \cap F} q$ for every $v \in F$, then $p R^F q$.

I take Comparative Deference on F to hold for every nonempty F .

Suppose that E_A and E_B satisfy Truth, and suppose A and B commonly know at w that A judges $p \succeq q$ and B judges $q \succ p$. Then there is a public F containing w throughout which A judges $p \succeq q$ and B judges $q \succ p$. Since F is public, $E_A(v) \subseteq F$ for every $v \in F$, so by Comparative Conditionalization A's ordering at v is $\succeq^{E_A(v)} = \succeq^{E_A(v) \cap F}$. Since A judges $p \succeq q$ throughout F , $p \succeq^{E_A(v) \cap F} q$ for every $v \in F$, and Comparative Deference on F with $\Pi = E_A$ gives $p \succeq^F q$. The same reasoning for B gives $q \succ^F p$. Contradiction.

6 BETTING AND DEFERENCE

Milgrom & Stokey (1982) show that if it is common knowledge that a trade is acceptable to everyone, then everyone is indifferent between trading and not trading.³⁰ Sebenius & Geanakoplos (1983) prove a similar result for a single bet: two people with a shared prior cannot commonly know that each strictly prefers her own side of the bet to declining it. Both results assume partitional evidence. Geanakoplos (1989, Corollary 5.1) shows that the Sebenius–Geanakoplos result survives when partitionality is replaced by positive balancedness. By the equivalence between positive balancedness and Total Containment, the no-betting result follows from Total Containment too.

6.1 The No-Betting Result

I represent a bet between A and B by a random variable X , where $X(w)$ is A's payoff at w and $-X(w)$ is B's. A *accepts* her side of the bet at w if $\mathbb{E}_{p_A(w)}[X] \geq 0$, and *strictly prefers her side* if $\mathbb{E}_{p_A(w)}[X] > 0$. B *accepts* her side at w if $\mathbb{E}_{p_B(w)}[-X] \geq 0$, and *strictly prefers her side* if $\mathbb{E}_{p_B(w)}[-X] > 0$.

Theorem 6.1. *Let $\mathcal{F}$ be a precise agreement frame satisfying the conditions of §2.1. Let X be a bet. Let $w \in \Omega$. Then A and B cannot commonly know at w that A strictly prefers X while B strictly prefers $-X$.³¹*

Proof. Suppose it is common knowledge at w that A strictly prefers X and B strictly prefers $-X$. Choose a public F containing w throughout which both preferences hold. By Conditionalization,

$$\mathbb{E}_\pi[X|E_A(v)] > 0 \quad \text{and} \quad \mathbb{E}_\pi[X|E_B(v)] < 0$$

for every $v \in F$.

Since A's evidence is partitional and F is self-evident to A, the distinct sets $E_A(v)$, for $v \in F$, partition F . By Truth, each is nonempty. By Regularity, each has positive probability. So $\pi(\cdot|F)$ is a weighted sum of the $\pi(\cdot|E_A(v))$, with nonnegative weights summing to one. By Lemma 2.8, $\mathbb{E}_\pi[X|F]$ is a weighted sum of the $\mathbb{E}_\pi[X|E_A(v)]$, with the same weights. It follows that $\mathbb{E}_\pi[X|F] > 0$. Applying the same reasoning to B, we conclude that $\mathbb{E}_\pi[X|F] < 0$. Contradiction. $\square$

30 Milgrom & Stokey assume a shared prior and that, before trading, there is no way to make anyone better off without making someone else worse off. Without the second assumption, a trade can benefit everyone, and those benefits can be common knowledge.

31 Sebenius & Geanakoplos (1983, Proposition 2) in fact prove a stronger result. Suppose it is common knowledge that A's expectation of X is no lower than a , while B's expectation of X is no higher than a . Sebenius & Geanakoplos show that both expectations must equal a , and that this equality is itself common knowledge.

6.2 Positive Balancedness and Total Containment

Geanakoplos shows:

Theorem 6.2. *Let $\mathcal{F}$ be a precise agreement frame satisfying the conditions of §2.1 except Positive and Negative Introspection. Suppose E_A and E_B are each positively balanced on every nonempty public proposition. Let X be a bet. Let $w \in \Omega$. Then A and B cannot commonly know at w that A strictly prefers X while B strictly prefers $-X$.*

Proof. The proof of Theorem 6.1 used partitionality only to write $\pi(\cdot|F)$ as a weighted sum of the $\pi(\cdot|E_A(v))$, and likewise of the $\pi(\cdot|E_B(v))$, with nonnegative weights summing to one. Positive balancedness on F supplies these weighted sums directly. The rest of the proof is unchanged. $\square$

We can also derive Theorem 6.2 using Total Containment, following §4.2. Suppose for contradiction that both strict preferences are commonly known at w , and choose a public F containing w throughout which both preferences hold. Since F is public, $E_A(v) \cap F = E_A(v)$ for every $v \in F$. A 's estimates $\mathbb{E}_\pi[X|E_A(v)]$ are strictly positive throughout F . Because F is finite, these estimates all lie in a closed interval $[a, a']$ with $a > 0$. Total Containment on F , with $\Pi = E_A$, therefore requires $\mathbb{E}_\pi[X|F] \in [a, a']$, so $\mathbb{E}_\pi[X|F] > 0$. Applying Total Containment to B 's negative estimates instead requires $\mathbb{E}_\pi[X|F] < 0$. Contradiction.

7 CONCLUSION

Each of the agreement theorems in this paper follows from a principle of deference. Total Certainty yields Aumann's theorem. Total Containment yields the Overlap Theorem, the no-betting result, and a result about precise estimates: those who satisfy it cannot commonly know that their estimates lie in disjoint intervals. Comparative Deference yields No Comparative Disagreement. Yet the theorems themselves say nothing about deference. They rely instead on structural conditions on evidence. Aumann assumed partitionality. Geanakoplos showed that balancedness suffices for agreement. I have shown that positive balancedness suffices for overlap.

Balancedness and positive balancedness are obscure conditions that seem to have no epistemic significance of their own. I have shown that each is equivalent to a principle of deference: balancedness to Total Certainty, and positive balancedness to Total Containment. The equivalences have a further consequence. Among those who share a prior, those who satisfy Total Certainty cannot agree to disagree with one another, and anyone who does not can agree to disagree with

someone who does. Likewise for Total Containment and agreeing to disagree in intervals.

The reason deference yields agreement is the same in each case. Suppose we share a prior and our estimates or comparative judgments are common knowledge. There is then a public proposition F throughout which those estimates or judgments are fixed. Conditional on F , my prior must defer to my better-informed self, and your prior must defer to yours. But my prior is your prior. So a single prior is bound by both constraints: our precise estimates must coincide, our imprecise estimates must overlap, and our rankings of p and q cannot conflict. In each case, the argument rests on the same three ingredients: a shared prior, conditionalization, and deference on public propositions.

The moral is that the agreement theorems are not as mysterious as they might seem. They follow from a simple and compelling thought—that we should defer to our better-informed selves.

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